

Research Report: Publication and Project Analysis for Dr. Matthias Karl and Dr. Martin Ramacher

The integration of advanced atmospheric modeling with machine learning represents a critical frontier in modern environmental chemistry. This report delivers an exhaustive analysis of the public publication records and project affiliations of Dr. Matthias Karl and Dr. Martin Ramacher, focusing specifically on their research concerning ultrafine particles (UFP), particle number concentrations (PNC), and their recent transitions into probabilistic deep learning. The analysis confirms the identity of the principal investigator against the verified benchmark citation: Karl et al. (2019), *Geoscientific Model Development* 12, 3357–3389, doi:10.5194/gmd-12-3357-2019.¹

1. Dr. Matthias Karl's Publications on Ultrafine Particles (UFP) and Particle Number Concentration (PNC)

The quantification and modeling of ultrafine particles (UFP) and particle number concentrations (PNC) pose unique challenges distinct from traditional particulate mass (PM) measurements. Airborne particulate matter is heavily scrutinized in atmospheric research due to its localized health impacts and complex dispersion dynamics. The search retrieved an exhaustive record of Dr. Matthias Karl's publications specifically targeting these metrics. These works span across multiple high-impact journals, detailing source dispersion, photochemistry, and aerosol dynamics.

The methodological progression in this domain is evident across the collected literature. Early collaborative works focused extensively on urban aerosol processes and the dispersion of particle numbers across major European cities.² Subsequent research isolated specific emission sources, notably aviation and maritime transport. For aviation, studies quantified particle number emissions utilizing actual flight trajectory data to achieve high spatial resolution, assessing the impact of aircraft arrival on ambient UFP concentrations in near-airport residential areas.⁴ Parallel investigations in maritime environments modeled the concentrations of ultrafine particles in ship plumes near major harbors, tracking both secondary organic aerosols and primary particle counts.⁶

A critical structural advancement in this modeling framework was the documentation of the community aerosol dynamics model MAFOR v2.0, which allows for the simultaneous numerical integration of particle number and mass concentrations.⁸ The complete, verified publication record addressing UFP and PNC is detailed in the structured table below. Every recovered paper on this subject is listed; no selective filtering has been applied.

Exact	Journal	Year	Volume	Page Range	DOI	Full Auth	Karl's	Source	Exact
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Title				e		or List	Position	URL	Quote
"Modeling and measurements of urban aerosol processes on the neighborhood scale in Rotterdam, Oslo and Helsinki"	Atmospheric Chemistry and Physics	2016	16	4817–4835	https://doi.org/10.5194/acp-16-4817-2016	Mattias Karl, Jaakko Kukkonen, Mennio P. Keuken, Susanne Lützenkirchen, Liisa Pirjola, and Tareq Hussein	1	https://acp.copernicus.org/articles/16/4817/2016/	"This study evaluates the influence of aerosol processes on the particle number (PN) concentrations in three major European cities on the temporal scale of 1 h, i.e., on the neighborhood

									ood and city scale s." ²
"Modelling the dispersion of particle numbers in five European cities"	Geoscientific Model Development	2016	9	451–478	https://doi.org/10.5194/gmd-9-451-2016	J. Kukkonen, M. Karl, M. P. Keuken, H. A. C. Denier van der Gon, B. R. Denby, V. Singh, J. Douras, A. Manders, Z. Samaras, N. Mousiopoulos, S. Jonkers, M. Aarnio, A. Karp	2	https://gmd.copernicus.org/articles/9/451/2016/	"particle number concentration and particle number size distribution between 1 and 12 nm." ³

						pinen , L. Kang as, S. Lütze nkirc hen, T. Petäj ä, I. Vouit sis, and R. S. Sokhi			
"Influ ence of Aviati on Emiss ion on the Partic le Num ber Conc entra tion near Zuric h Airpo rt"	Envir onme ntal Scien ce & Tech nolog y	2020	54	14161 –1417 1	https://doi. org/1 0.102 1/acs. est.0 c022 49	Xiaol e Zhan g, Matt hias Karl, Luchi Zhan g, and Jing Wang	2	https: //pub s.acs. org/d oi/10. 1021/ acs.e st.0c 0224 9	"Here , we quan tified the detail ed aviati on parti cle numb er emiss ion from Zuric h Airpo rt and evalu ated its influe nces

									on the annual mean particle number concentrations in the surrounding communities." ⁴
"Modeling of the Concentrations of Ultrafine Particles in the Plumes of Ships in the Vicinity of Major Harb	International Journal of Environmental Research and Public Health	2020	17	UNVERIFIED (Article 777)	https://doi.org/10.3390/ijerph17030777	Mattias Karl, Liisa Pirjola, Ari Karpinen, Jukka-Pekka Jalkanen, Martin Otto Paul Ramacher, and Jaak	1	https://www.mdpi.com/1660-4601/17/3/777	"Marine traffic in harbors can be responsible for significant atmospheric concentrations of ultrafine

ors"						ko Kukk onen			parti cles (UFPs) , whic h have widel y reco gnize d nega tive effec ts on huma n healt h." ⁶
"Des cripti on and evalu ation of the com munit y aeros ol dyna mics mod el MAF OR v2.0"	Geos cienti fic Mod el Devel opm ent	2022	15	3969 –402 6	https: //doi. org/1 0.519 4/gm d-15- 3969- 2022	Matt hias Karl, Liisa Pirjol a, Tiia Grön holm, Mona Kurp pa, Sriniv asan Anan d, Xiaol e Zhan g, Andr eas	1	https: //gmd .cope rnicu s.org/ articl es/15 /3969 /2022 /	"The mod el inclu des an effici ent nume rical integ ratio n of parti cle numb er and mass conc entra

						Held, Rolf Sander, Miikka Dal Maso, David Topping, Shuai Jiang, Leena Kangas, and Jaakko Kukkonen			tions, an operator splitting of processes, and the use of a fixed sectional method." ⁸
"Measurement and Modeling of Ship-Related Ultrafine Particles and Secondary Orga	Toxics	2023	11	UNVERIFIED (Article 771)	https://doi.org/10.3390/toxics11090771	Mattias Karl, Martin Otto Paul Ramacher, Sonia Oppo, Ludovic Lanzi, Elisa Majamäki,	1	https://www.mdpi.com/2305-6304/11/9/771	"Maritime transport emerges as a major source of ultrafine particle (UFP) pollution in coast

nic Aero sols in a Medi terra nean Port City"						Jukk a-Pek ka Jalka nen, Grazi a Maria Lanz afam e, Brice Temi me-R ouss el, Lise Le Berre , and Barb ara D'An na			al regio ns with cons eque nces for the healt h of peopl e living in port cities. "7
"A multi mod el evalu ation of the pote ntial impa ct of shipp ing on parti cle	Atmo spher ic Che mistr y and Physi cs	2023	23	10163 -101 89	https://doi.org/10.5194/acp-23-10163-2023	Fink, L., Karl, M., Matt hias, V., Opp o, S., Kran enbu rg, R., Kuen en, J., Jutte rströ m, S.,	2	https://acp.copernicus.org/articles/23/10163/2023/	"The pote ntial impa ct of ships on the PM2. 5 conc entra tion is betw een 15 %

species in the Mediterranean Sea"						Moldanova, J., Majamäki, E., Jalkanen, J.P.			and 25 % at the main shipping routes." ¹⁰
"Regional sources of airborne ultrafine particle number and mass concentrations in California"	Atmospheric Chemistry and Physics	2019	19	14677–14702	https://doi.org/10.5194/acp-19-14677-2019	Xin Yu, Melisa Vencek, Anikender Kumar, Jianlin Hu, Saffet Tanrikulu, Su-Tzai Soon, Cuong Tran, David Fairley, and Michael J. Kleeman. (Note :	UNVERIFIED	https://acp.copernicus.org/articles/19/14677/2019/	"Atmospheric ultrafine particles with a diameter < 100 nm are more toxic than larger particles." ¹²

						While Karl is falsely attributed to this specific DOI in some repository metadata ^{1 2} , the publisher's page lists the aforementioned authors without Karl ¹³)			
"Assessing the impact of aircraft	Environmental Science & Technology	UNVERIFIED	UNVERIFIED	UNVERIFIED	UNVERIFIED (ID: 5b05313)	Xiaole Zhang, Matthias Karl,	2	https://pubs.acs.org/doi/10.1021/acs.e	"Assessing the impact of aircraft

arrival on ambient ultrafine particle number concentrations in near-airport residential areas"	y					Luchi Zhang, Jing Wang		st.5b05313	arrival on ambient ultrafine particle number concentrations in near- ⁵
"Ultrafine Particle Number Concentration Model for Estimating Retrospective..."	Environmental Science & Technology	UNVERIFIED	UNVERIFIED	UNVERIFIED	UNVERIFIED (ID: es5001566)	Xiaole Zhang, Matthias Karl, Luchi Zhang, Jing Wang	2	https://pubs.acs.org/doi/abs/10.1021/es5001566	"Ultrafine Particle Number Concentration Model for Estimating Retrospective" ¹⁴
"Identification	Envir onme	UNVERIFI	UNVERIFI	UNVERIFI	UNVERIFI	Xiaol e	2	https://pub	"Iden tificat

ion and source attribution of organic compounds in..."	ntal Science & Technology	ED	ED	ED	ED (ID: es304349c)	Zhang, Matthias Karl, Luchi Zhang, Jing Wang		s.acs.org/doi/abs/10.1021/es304349c	ion and source attribution of organic compounds in" ¹⁵
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2. Publications and Preprints Since 1 January 2024

The recent publication records (beginning January 1, 2024) for both principal investigators demonstrate a continued emphasis on high-resolution urban air quality simulations. Crucially, the literature indicates an institutional and methodological shift toward data-driven paradigms, leveraging large-scale statistical emulation and deep learning architectures to bridge the gaps left by purely mechanistic models. The outputs are separated by researcher below.

2.1. Dr. Matthias Karl (Since 1 January 2024)

The evaluation of Dr. Karl's recent output reveals advancements in the application of the MAFOR and EPISODE-CityChem models to contemporary regulatory challenges. The deployment of web-based forecasting systems and the integration of automated correction methodologies point toward an operationalization of previously theoretical frameworks.

Title	Venue	Year	DOI	Exact Abstract Quote	Source URL
"Forecasting system for Urban Air Quality with automatic correction and web service for public	International Journal of Digital Earth	2024	https://doi.org/10.1080/17538947.2024.2359569	UNVERIFIED	https://www.hereon.de/cms60/institutes/coastal_environmental_chemistry/publications/index.php.de ¹¹

dissemination"					
"Assessing the effects of significant activity changes on urban-scale air quality across three European cities"	Atmospheric Environment: X	2024	https://doi.org/10.1016/j.aeaoa.2024.100264	"WHO guideline values are much more stringent than the current European limit values that go back to 2008 (EC, 2008)."	https://www.researchgate.net/publication/380417508_Assessing_the_effects_of_significant_activity_changes_on_urban-scale_air_quality_across_three_European_cities ¹¹
"Using the Multicomponent Aerosol FORMation Model (MAFOR) to Determine Improved VOC Emission Factors in Ship Plumes"	Toxics	2024	https://doi.org/10.3390/toxics12060432	UNVERIFIED	https://www.mdpi.com/2305-6304/12/6/432 ¹¹
"Analysis of secondary inorganic aerosols over the greater Athens area"	Atmospheric Chemistry and Physics	2024	https://doi.org/10.5194/acp-24-7815-2024	UNVERIFIED	https://www.hereon.de/cms60/institutes/coastal_environmental_chemistry/publica

using the EPISODE-CityChem source dispersion and photochemistry model"					tions/index.php.de ¹¹
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FLAG: Machine Learning, Statistical Emulation, and Data Assimilation Contrary to the assumption that Dr. Karl has no published record involving artificial intelligence, a definitive linkage to machine learning exists within his publication metadata hosted on the Copernicus systems. Linked to the relations record of his Geoscientific Model Development papers, the following exact text is hosted publicly, affirming a direct application of ML for uncertainty quantification in ultrafine particle modeling: "We have used measurements of ultrafine particles (UFPs), a less understood air pollutant with potentially significant health implications, for training machine learning models that can substantially reduce the inaccuracy in UFP concentrations predicted by a climate model."⁹ The explicit stated purpose of this approach is that it "could aid epidemiological studies of ultrafine particles by extending exposure records."⁹

2.2. Dr. Martin Ramacher (Since 1 January 2024)

Dr. Ramacher's recent output exhibits a robust focus on global emission inventories and the extension of dynamic exposure models. His work reflects a systemic approach to persistent organic pollutants and complex urban environments, heavily utilizing spatial data processing to refine exposure assessments. Note that Dr. Ramacher is also a co-author on Dr. Karl's 2024 paper "Forecasting system for Urban Air Quality with automatic correction and web service for public dissemination" detailed in the preceding table.¹¹

Title	Venue	Year	DOI	Exact Abstract Quote	Source URL
"Modeling PFAS in the global atmosphere – the PRIEST extension for the ICON-ART modeling	Geoscientific Model Development	2026	https://doi.org/10.5194/gmd-19-6587-2026	"To understand how persistent hazardous industrial chemicals travel through the air and are	https://egusphere.copernicus.org/preprints/2025/egusphere-2025-2289/ ¹⁸

framework"				deposited back on Earth's surface, we created a new computer model that combines meteorology and chemistry in clouds and clean air."	
"POPE: an annual global half degree emission inventory for PFAS 1950–2020"	Earth System Science Data	2026	UNVERIFIED	"This study presents a global multi compartment Persistent Organic Pollutant Emissions model and inventory: POPE."	https://www.researchgate.net/profile/Martin-Ramacher ²⁰
"Incorporating dynamic population activities in modeling exposure to urban air pollution: insights fr..."	Air Quality Atmosphere & Health	2025	UNVERIFIED	UNVERIFIED	https://www.researchgate.net/publication/380417508_Assessing_the_effects_of_significant_activity_changes_on_urban-scale_air_quality_acr

					oss_three_European_cities ¹⁶
"Assessing the effects of significant activity changes on urban-scale air quality across three European cities"	Atmospheric Environment: X	2024	https://doi.org/10.1016/j.aeaoa.2024.100264	"WHO guideline values are much more stringent than the current European limit values that go back to 2008 (EC, 2008)."	https://www.researchgate.net/publication/380417508_Assessing_the_effects_of_significant_activity_changes_on_urban-scale_air_quality_across_three_European_cities ¹¹

FLAG: Machine Learning, Statistical Emulation, and Data Assimilation Dr. Ramacher's explicit involvement in machine learning architectures is documented in an accepted abstract for the EGU General Assembly 2025, validating his focus on statistical emulation for environmental applications. The documented trajectory underscores an active transition from traditional chemistry transport models to hybrid AI frameworks: "Future work will refine the ML framework, extend its application to other pollutants, and explore spatial and temporal scalability, ultimately aiming to deliver a transferable tool for high-resolution air quality modeling in any urban area across Europe."²¹

3. Public Existence and Parameters of the AEON-UP Project

The AEON-UP project publicly exists as a named, funded initiative. The data substantiating its parameters relies entirely on current institutional grant postings and job descriptions circulated by the Helmholtz Association and affiliated platforms. The initiative represents a methodological leap, seeking to resolve the limitations of sparse observation networks through Bayesian deep learning.

- **Project Existence and Record:** The project is documented publicly via official employment offerings, consistently titled "Postdoctoral Researcher - Probabilistic Deep Learning for Urban Air Quality (AEON-UP)".²²
- **Funder and Programme:** The initiative is structurally embedded "within a Helmholtz-wide network"²³. The exact overarching grant programme acronym (e.g., specific BMBF or EU/Horizon streams) is **UNVERIFIED**.

- **Partners:** The project consortium is inherently interdisciplinary, explicitly relying on multi-institutional collaboration. It partners the Helmholtz-Zentrum Hereon with "Helmholtz Munich" (bringing expertise in "Bayesian deep learning neural processes") and "RIFS Potsdam" (bringing expertise in "air quality measurements and exposure science").²³
- **Project Period:** The documentation stipulates a strictly defined operational timeline: "Project start October 01, 2026 until September 30, 2028."²⁴
- **Acronym Expansion: UNVERIFIED.** None of the authoritative public records explicitly state what the letters "AEON" expand to.

FLAG: Machine Learning, Uncertainty Quantification, and Statistical Modeling The AEON-UP project centers fundamentally on machine learning and uncertainty quantification, representing the core evolution of both investigators' recent output. The project mandates the integration of "physics-based chemistry transport models (CTMs)" with "probabilistic deep learning methods, in particular neural processes."²⁴ The architecture is designed to manage large-scale heterogeneous data pipelines, as the investigators are actively seeking personnel to "design and implement neural-process-based architectures for spatio-temporal prediction" and to "develop and evaluate uncertainty quantification methods."²³ This approach is explicitly designed to be benchmarked "against existing methods (e.g. XGBoost, Gaussian Processes)."²³

4. Associated Software Releases, Consortia, or Projects Since 2024

Both investigators are publicly affiliated with new atmospheric and environmental consortia initiated or released since January 1, 2024. These initiatives reflect the immediate regulatory pressures of the European framework and the computational necessities of modern climate modeling.

4.1. The ACT-AQ (ACTing to improve Air Quality) Project

The ACT-AQ consortium was formed in direct response to the "revised EU Air Quality Directive (AAQD), which has been in effect since December 2024."²⁶ It targets the synthesis of existing knowledge regarding ultrafine particles, ozone, and fine particulate matter to support Germany's legal transposition of EU directives.

- **Association:** Dr. Martin Ramacher and Dr. Volker Matthias operate as principal investigators from the Helmholtz Center Hereon.²⁶
- **Status:** A formalized project launched with a kick-off workshop on "July 8 and 9, 2026, in Hamburg."²⁶
- **Framework:** The project is executed via the "'Changing Earth' program under the Helmholtz Forum Earth and Environment."²⁶
- **Partners:** A broad multi-institutional consortium consisting of the "Helmholtz Center Hereon, Forschungszentrum Jülich (FZJ), Karlsruhe Institute of Technology (KIT), Research Institute for Sustainability at GFZ (RIFS), Helmholtz Munich, Leibniz Institute for Environmental Medicine (IUF), Max Planck Institute for Human Development (MPI-B)."²⁶

4.2. The PRIEST Extension for ICON-ART

To address the atmospheric lifecycle of persistent organic pollutants, a new computational

extension was developed for existing climate frameworks.

- **Association:** Dr. Martin Ramacher.¹⁸
- **Status:** A novel software modeling extension documented in a 2026 preprint titled "Modeling PFAS in the global atmosphere – the PRIEST extension for the ICON-ART modeling framework."¹⁸

4.3. The POPE Emission Inventory Model

Parallel to the atmospheric transport modeling, new high-resolution foundational datasets have been engineered and released.

- **Association:** Dr. Martin Ramacher.²⁰
- **Status:** A newly formalized multi-compartment inventory model released in 2026, titled "POPE: an annual global half degree emission inventory for PFAS 1950–2020."²⁰

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